Linux SATA Developer Guide

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Preface

Overview

Product Version

Chipset	Kernel Version
RK356X	4.19
RK3588/RK3588S/RK356X	5.10
RK3576	6.1

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Zhao Yifeng	2021-02-26	Initial version
V1.1.0	Zhao Yifeng	2021-04-30	Added ACT LED configuration
V2.0.0	Zhao Yifeng	2021-12-23	Added RK3588 support Added speed limit configuration information Corrected RK3566 support SATA information
V2.1.0	Zhao Yifeng	2022-09-08	Added JMB575 firmware download instructions Added support for hard disk quantity limit instructions Added U-boot support instructions
V2.2.0	Zhao Yifeng	2023-07-20	Supplemented U-boot DTS configuration instructions Added sleep and standby settings instructions
V2.3.0	Zhao Yifeng	2024-04-28	Added RK3576 support
V2.4.0	Zhao Yifeng	2025-04-22	Added kernel 6.1 configuration

Contents

Linux SATA Developer Guide

1. Chip Resource Introduction
2. DTS Configuration
3. menuconfig Configuration
4. U-boot Support for SATA
   - 4.1 DTS Modification
   - 4.2 Compilation
5. Common Issues
   - 5.1 Does it support downloading firmware to PM chips via SATA interface?
   - 5.2 How to Test SATA Performance?
   - 5.3 Why does the default SDK code not recognize SATA devices?
   - 5.4 How to Limit SATA Speed to 1.5G or 3G?
   - 5.5 How to Enable FBS Feature on RK356X?
   - 5.6 How to Set Hard Disk to Enter Standby or Sleep Mode?

1. Chip Resource Introduction

RK3566

Resource	Mode	Support for PM Chip Expansion	PHY Multiplexing	Remarks
SATA1	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	USB, QSGMII	
SATA2	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE, QSGMII	

RK3568

Resource	Rate	Support for PM Chip Expansion	PHY Multiplexing	Remarks
SATA0	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	USB	
SATA1	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	USB, QSGMII	
SATA2	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE, QSGMII	

RK3588S

Resource	Mode	Support for PM Chip Expansion	PHY Multiplexing	Remarks
SATA0	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE	
SATA2	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE, USB	

RK3588

Resource	Rate	Support for PM Chip Expansion	PHY Multiplexing	Remarks
SATA0	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE	
SATA1	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE	
SATA2	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE, USB	

RK3576

Resource	Rate	Support for PM Chip Expansion	PHY Multiplexing	Remarks
SATA0	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE	
SATA1	6G/3G/1.5G	Supported (Supports 5 Hard Disks)	PCIE, USB	

2. DTS Configuration

RK3566

Resource	Reference Configuration	Controller Node	PHY Node
SATA1	rk3568.dtsi	sata1	combphy1_usq
SATA2	rk3568.dtsi	sata2	combphy2_psq

RK3568

Resource	Reference Configuration	Controller Node	PHY Node
SATA0	rk3568.dtsi	sata0	combphy0_us
SATA1	rk3568.dtsi	sata1	combphy1_usq
SATA2	rk3568.dtsi	sata2	combphy2_psq

RK3588

Resource	Reference Configuration	Controller Node	PHY Node
SATA0	rk3588s.dtsi	sata0	combphy0_ps
SATA1	rk3588.dtsi	sata1	combphy1_ps
SATA2	rk3588s.dtsi	sata2	combphy2_psu

RK3588s

Resource	Reference Configuration	Controller Node	PHY Node
SATA0	rk3588s.dtsi	sata0	combphy0_ps
SATA2	rk3588s.dtsi	sata2	combphy2_psu

RK3576

Resource	Reference Configuration	Controller Node	PHY Node
SATA0	rk3576.dtsi	sata0	combphy0_ps
SATA1	rk3576.dtsi	sata1	combphy1_psu

1. `compatible = "rockchip,rk-ahci", "snps,dwc-ahci";`

Mandatory Configuration: Default configuration, corresponding driver: drivers/ata/ahci_platform.c.

2. `phy-names = "sata-phy";`

Mandatory Configuration: Cannot be modified, AHCI driver will find the corresponding combphy node based on the "sata-phy" name.

3. `status = <okay>;`

Mandatory Configuration: This configuration needs to be enabled in both the SATA controller node and the corresponding phy node.

4. `assigned-clock-rates = <100000000>;`

Mandatory Configuration: The reference clock frequency values that can be configured are: 24000000, 25000000, and 100000000, with 100000000 being the recommended configuration.

5. `pinctrl-0 = <&sata_pm_reset>;`

```
sata_pm_reset: sata-pm-reset {
    rockchip,pins = <4 RK_PD2 RK_FUNC_GPIO &pcfg_output_high>;
};

&combphy1_usq {
    pinctrl-names = "default";
    pinctrl-0 = <&sata_pm_reset>;
};
```

Optional Configuration: When extending the SATA port with an external PM chip, a GPIO may be required to reset the PM chip. For specific settings, refer to "rk3568-nvr-demo-v10.dtsi".

6. `pinctrl-0 = <&sata2_pins>;`

```
&sata0 {
    pinctrl-names = "default";
    pinctrl-0 = <&sata0_pins>;
};

&sata1 {
    pinctrl-names = "default";
    pinctrl-0 = <&sata1_pins>;
};

&sata2 {
    pinctrl-names = "default";
    pinctrl-0 = <&sata2_pins>;
};
```

Optional Configuration: Add pinctrl configuration to the corresponding sata controller node to enable the SATA ACT LED function.

6. `dma-coherent;`

Optional Configuration: RK3576 supports CACHE coherency, and the SDK has enabled this feature by default. The dts can be configured, but other platforms cannot configure this parameter.

3. menuconfig Configuration

Ensure the following configurations are enabled to correctly use SATA-related functions.

For kernel 4.19 and kernel 5.10, the configurations are as follows:

```
CONFIG_ATA=y
CONFIG_SATA_AHCI=y
CONFIG_SATA_AHCI_PLATFORM=y
CONFIG_PHY_ROCKCHIP_NANENG_COMBO_PHY=y
```

For kernel 6.1, the configurations are as follows:

```
CONFIG_ATA=y
CONFIG_SATA_AHCI=y
CONFIG_AHCI_DWC=y
CONFIG_PHY_ROCKCHIP_NANENG_COMBO_PHY=y
```

4. U-boot Support for SATA

4.1 DTS Modification

DTS requires enabling the combphy and SATA controllers, and the referenced php_grf and pipe_phyx_grf nodes must also be enabled. Below is a configuration reference for RK3588 SATA0:

```
diff --git a/arch/arm/dts/rk3588-u-boot.dtsi b/arch/arm/dts/rk3588-u-boot.dtsi
index 3fe8054aac..29cb50e122 100644
--- a/arch/arm/dts/rk3588-u-boot.dtsi
+++ b/arch/arm/dts/rk3588-u-boot.dtsi
@@ -192,6 +192,16 @@
        status = "okay";
 };

+&sata0 {
+       u-boot,dm-pre-reloc;
+       status = "okay";
+};
+
+&combphy0_ps {
+       u-boot,dm-pre-reloc;
```

```
+       status = "okay";
+};
+
 /* Support SPL-PINCTRL:
  * 1. ioc
  * 2. pinctrl(sdmmc)
```

4.2 Compilation

The configuration file for RK356x is "rk3568-sata.config", and the compilation command is:

```
./make.sh rk3568-sata
```

The configuration file for RK3588(s) is "rk3588-sata.config", and the compilation command is:

```
./make.sh rk3588-sata
```

5. Common Issues

5.1 Does it support downloading firmware to PM chips via SATA interface?

Both RK356X and RK3588 support firmware downloading using JMB575, patch download address: JMB575 Firmware Download Patch

5.2 How to Test SATA Performance?

5.3 Why does the default SDK code not recognize SATA devices?

Software Situation 1: After updating uboot separately with the first version of the SDK code, this issue may occur, requiring an update to the phy driver under the kernel.

```
commit b3f78165e536d35b2337063093bb33a018ff518d
Author: Shawn Lin <shawn.lin@rock-chips.com>
Date:   Wed Dec 23 16:17:31 2020 +0800

    phy: rockchip: naneng-combphy: Reset phy if not being used

    Change-Id: Ia62481ebf5aa5684c359fd00a3933bb02e2caaff
    Signed-off-by: Shawn Lin <shawn.lin@rock-chips.com>
```

Software Situation 2: SATA and PCIE can only be chosen one at a time, and the PCIE should be configured as disabled in the DTS.

Hardware Situation: SATA and PCIE can only be chosen one at a time. If a compatible layout is implemented, it will affect the signal. Remove the branch of the PCIE wiring part.

5.4 How to Limit SATA Speed to 1.5G or 3G?

The PHY driver defaults to supporting 6G speeds. By modifying the PHY driver, the highest supported rate of the PHY can be limited. Below are reference patches for different platforms.

RK356X Patch to Limit to 1.5G:

```
diff --git a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
index 08445c1890eb..3df0e0e05ab4 100644
--- a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
+++ b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
@@ -604,7 +604,7 @@ static const struct rockchip_combphy_grfcfg
rk3568_combphy_grfcfgs = {
        .con2_for_sata          = { 0x0008, 15, 0, 0x00, 0x80c3 },
        .con3_for_sata          = { 0x000c, 15, 0, 0x00, 0x4407 },
        /* pipe-grf */
-       .pipe_con0_for_sata     = { 0x0000, 15, 0, 0x00, 0x2220 },
+       .pipe_con0_for_sata     = { 0x0000, 15, 0, 0x00, 0x0000 },
        .pipe_sgmii_mac_sel     = { 0x0040, 1, 1, 0x00, 0x01 },
        .pipe_xpcs_phy_ready    = { 0x0040, 2, 2, 0x00, 0x01 },
        .u3otg0_port_en         = { 0x0104, 15, 0, 0x0181, 0x1100 },
```

RK356X Patch to Limit to 3G:

```
diff --git a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
index 08445c1890eb..3df0e0e05ab4 100644
--- a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
+++ b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
@@ -604,7 +604,7 @@ static const struct rockchip_combphy_grfcfg
rk3568_combphy_grfcfgs = {
        .con2_for_sata          = { 0x0008, 15, 0, 0x00, 0x80c3 },
        .con3_for_sata          = { 0x000c, 15, 0, 0x00, 0x4407 },
        /* pipe-grf */
-       .pipe_con0_for_sata     = { 0x0000, 15, 0, 0x00, 0x2220 },
+       .pipe_con0_for_sata     = { 0x0000, 15, 0, 0x00, 0x1110 },
        .pipe_sgmii_mac_sel     = { 0x0040, 1, 1, 0x00, 0x01 },
        .pipe_xpcs_phy_ready    = { 0x0040, 2, 2, 0x00, 0x01 },
        .u3otg0_port_en         = { 0x0104, 15, 0, 0x0181, 0x1100 },
```

RK3588(S) Patch to Limit to 1.5G:

```
diff --git a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
index 6f7aa8ebc292..062116ad09e8 100644
--- a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
+++ b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
@@ -868,8 +868,8 @@ static const struct rockchip_combphy_grfcfg
rk3588_combphy_grfcfgs = {
        .con2_for_sata          = { 0x0008, 15, 0, 0x00, 0x80c1 },
```

```
        .con3_for_sata          = { 0x000c, 15, 0, 0x00, 0x0407 },
        /* pipe-grf */

-       .pipe_con0_for_sata     = { 0x0000, 11, 5, 0x00, 0x22 },
-       .pipe_con1_for_sata     = { 0x0004, 2, 0, 0x00, 0x2 },

+       .pipe_con0_for_sata     = { 0x0000, 11, 5, 0x00, 0x00 },
+       .pipe_con1_for_sata     = { 0x0004, 2, 0, 0x00, 0x0 },
         };

 static const struct clk_bulk_data rk3588_clks[] = {
```

RK3588(S) Patch to Limit to 3G:

```
diff --git a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
index 6f7aa8ebc292..062116ad09e8 100644
--- a/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
+++ b/drivers/phy/rockchip/phy-rockchip-naneng-combphy.c
@@ -868,8 +868,8 @@ static const struct rockchip_combphy_grfcfg
rk3588_combphy_grfcfgs = {
        .con2_for_sata          = { 0x0008, 15, 0, 0x00, 0x80c1 },
        .con3_for_sata          = { 0x000c, 15, 0, 0x00, 0x0407 },
        /* pipe-grf */

-       .pipe_con0_for_sata     = { 0x0000, 11, 5, 0x00, 0x22 },
-       .pipe_con1_for_sata     = { 0x0004, 2, 0, 0x00, 0x2 },

+       .pipe_con0_for_sata     = { 0x0000, 11, 5, 0x00, 0x11 },
+       .pipe_con1_for_sata     = { 0x0004, 2, 0, 0x00, 0x1 },
         };

 static const struct clk_bulk_data rk3588_clks[] = {
```

5.5 How to Enable FBS Feature on RK356X?

When using an external PM chip, enabling the FBS feature can improve read and write performance.

For kernel 4.19 and kernel 5.10, update the following patches to modify the SATA node in the dtsi to `compatible = "rockchip,rk-ahci", "snps,dwc-ahci";`

```
diff --git a/drivers/ata/ahci_platform.c b/drivers/ata/ahci_platform.c
index 3aab2e3d57f3..1825b33cc274 100644
--- a/drivers/ata/ahci_platform.c
+++ b/drivers/ata/ahci_platform.c
@@ -62,6 +62,9 @@ static int ahci_probe(struct platform_device *pdev)
        if (of_device_is_compatible(dev->of_node, "hisilicon,hisi-ahci"))
                hpriv->flags |= AHCI_HFLAG_NO_FBS | AHCI_HFLAG_NO_NCQ;

+       if (of_device_is_compatible(dev->of_node, "rockchip,rk-ahci"))
+               hpriv->flags |= AHCI_HFLAG_YES_FBS;
+
        port = acpi_device_get_match_data(dev);
        if (!port)
```

```
                port = &ahci_port_info;
@@ -88,6 +91,7 @@ static const struct of_device_id ahci_of_match[] = {
        { .compatible = "snps,dwc-ahci", },
        { .compatible = "hisilicon,hisi-ahci", },
        { .compatible = "cavium,octeon-7130-ahci", },
+       { .compatible = "rockchip,rk-ahci", },
        {},
 };
```

For kernel 6.1, a different driver is used, and the following patch can be updated through u-boot to achieve this.

RK356X platform u-boot patch:

```
diff --git a/arch/arm/mach-rockchip/rk3568/rk3568.c b/arch/arm/mach-
rockchip/rk3568/rk3568.c
index 3b33322a6c7..89e0eb3ca5a 100644
--- a/arch/arm/mach-rockchip/rk3568/rk3568.c
+++ b/arch/arm/mach-rockchip/rk3568/rk3568.c
@@ -61,6 +61,13 @@ DECLARE_GLOBAL_DATA_PTR;

 #define EBC_PRIORITY_REG       (0xfe158008)

+#define SATA0_BASE_ADDR                        0xfc000000
+#define SATA1_BASE_ADDR                        0xfc400000
+#define SATA2_BASE_ADDR                        0xfc800000
+#define SATA_PI                                0xC
+#define SATA_PORT_CMD                  0x118
+#define SATA_FBS_ENABLE                        BIT(22)
+
 enum {
        /* PMU_GRF_GPIO0C_IOMUX_L */
        GPIO0C1_SHIFT           = 4,
@@ -945,6 +952,15 @@ int arch_cpu_init(void)
        writel((0x77771111), GRF_BASE + GRF_GPIO1C_IOMUX_L);
        writel((0x07770111), GRF_BASE + GRF_GPIO1C_IOMUX_H);
 #endif
+       /*
+        * Set SATA FBSCP and PORTS_IMPL for kernel drivers
+        */
+       writel(SATA_FBS_ENABLE, SATA0_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA0_BASE_ADDR + SATA_PI);
+       writel(SATA_FBS_ENABLE, SATA1_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA1_BASE_ADDR + SATA_PI);
+       writel(SATA_FBS_ENABLE, SATA2_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA2_BASE_ADDR + SATA_PI);
 #endif

        return 0;
```

RK3576 platform u-boot patch:

```
diff --git a/arch/arm/mach-rockchip/rk3576/rk3576.c b/arch/arm/mach-
rockchip/rk3576/rk3576.c
index c664c743f36..5758af56b13 100644
--- a/arch/arm/mach-rockchip/rk3576/rk3576.c
+++ b/arch/arm/mach-rockchip/rk3576/rk3576.c
@@ -94,6 +94,12 @@ DECLARE_GLOBAL_DATA_PTR;
```

```
 #define PMU1_CRU_GATE_CON03    0x080C
 #define PMU1_CRU_SOFTRST_CON03 0x0a0C

+#define SATA0_BASE_ADDR                         0x2a240000
+#define SATA1_BASE_ADDR                         0x2a250000
+#define SATA_PI                                 0xC
+#define SATA_PORT_CMD                   0x118
+#define SATA_FBS_ENABLE                         BIT(22)
+
 #ifdef CONFIG_ARM64
 #include <asm/armv8/mmu.h>

@@ -400,6 +406,14 @@ int arch_cpu_init(void)
          * Module: GMAC0/1, MMU0/1(PCIe, SATA, USB3)
          */
         writel(0xffffff00, SYS_SGRF_BASE + SYS_SGRF_SOC_CON20);
+#else
+       /*
+        * Set SATA FBSCP and PORTS_IMPL for kernel drivers
+        */
+       writel(SATA_FBS_ENABLE, SATA0_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA0_BASE_ADDR + SATA_PI);
+       writel(SATA_FBS_ENABLE, SATA1_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA1_BASE_ADDR + SATA_PI);
 #endif

 #if defined(CONFIG_ROCKCHIP_EMMC_IOMUX)
```

RK3588 platform u-boot patch:

```
diff --git a/arch/arm/mach-rockchip/rk3588/rk3588.c b/arch/arm/mach-
rockchip/rk3588/rk3588.c
index e42a1aef996..986bf56519b 100644
--- a/arch/arm/mach-rockchip/rk3588/rk3588.c
+++ b/arch/arm/mach-rockchip/rk3588/rk3588.c
@@ -102,6 +102,13 @@ DECLARE_GLOBAL_DATA_PTR;
 #define MMU600PHP_TCU_PRIORITY_REG     0xfdf3a808
 #define QOS_PRIORITY_LEVEL(h, l)       ((((h) & 7) << 8) | ((l) & 7))

+#define SATA0_BASE_ADDR                         0xfe210000
+#define SATA1_BASE_ADDR                         0xfe220000
+#define SATA2_BASE_ADDR                         0xfe230000
+#define SATA_PI                                 0xC
+#define SATA_PORT_CMD                   0x118
+#define SATA_FBS_ENABLE                         BIT(22)
+
 #ifdef CONFIG_ARM64
 #include <asm/armv8/mmu.h>

@@ -1067,6 +1074,16 @@ int arch_cpu_init(void)
          */
         writel(QOS_PRIORITY_LEVEL(4, 4), MMU600PHP_TBU_PRIORITY_REG);
         writel(QOS_PRIORITY_LEVEL(4, 4), MMU600PHP_TCU_PRIORITY_REG);
+
+       /*
+        * Set SATA FBSCP and PORTS_IMPL for kernel drivers
+        */
```

```
+       writel(SATA_FBS_ENABLE, SATA0_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA0_BASE_ADDR + SATA_PI);
+       writel(SATA_FBS_ENABLE, SATA1_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA1_BASE_ADDR + SATA_PI);
+       writel(SATA_FBS_ENABLE, SATA2_BASE_ADDR + SATA_PORT_CMD);
+       writel(1, SATA2_BASE_ADDR + SATA_PI);
 #endif

        /* Select usb otg0 phy status to 0 that make rockusb can work at high-
speed */
```

5.6 How to Set Hard Disk to Enter Standby or Sleep Mode?

When shutting down the system, it is necessary to set the hard disk to enter standby or sleep mode to prevent damage caused by direct power-off.

After executing the following commands, it may take about 10 seconds for the hard disk to spin down and enter standby or sleep mode. The hard disk will be re-awakened when accessed. It is recommended to unmount the hard disk before executing the command during shutdown to avoid system read/write access.

```
Standby: busybox hdparm -y /dev/block/sda
Sleep: busybox hdparm -Y /dev/block/sda
```